Regret-Bounded Combinatorial Bandits with Unreliable LLM Action Oracles

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Abstract

Combinatorial semi-bandits over d base arms with superarms of size m suffer a fundamental curse of dimensionality: standard CUCB achieves $\tilde{O}(\sqrt{md \cdot T})$ regret while the exponential action space makes full enumeration intractable for large d. Large language models can generate combinatorially structured suggestions reflecting world-knowledge priors unavailable to the bandit algorithm, yet these suggestions are systematically unreliable. We introduce a formal theory of *action-space corruption* in combinatorial bandits—qualitatively different from the reward-corruption model of Lykouris et al. [13]—where an external oracle corrupts the *action proposal* rather than the *reward observation*. We define the *Combinatorial LLM Oracle* (CLO) with a graded quality model parameterized by overlap fraction, corruption rate, and consistency, and propose **LLM-CUCB-AT**, an algorithm that adaptively trusts or hedges against the oracle using a composite trust score combining inter-query consistency with posterior validation against empirical estimates. We prove: (i) a gap-dependent regret bound showing dimension reduction from d to the oracle's effective support when the oracle is reliable; (ii) graceful degradation to vanilla CUCB rates under adversarial corruption, including against *consistently wrong* oracles; (iii) a matching information-theoretic lower bound for the action-corruption setting establishing that the corruption cost $\Omega(\varepsilon m/\Delta_{\min})$ is unavoidable; and (iv) an $O(\sqrt{T})$-query schedule reducing LLM cost from $O(T)$ to $O(\sqrt{T})$ with bounded regret increase. Experiments on synthetic semi-bandits, MIND news recommendation, and SNAP influence maximization validate the theoretical predictions.

1 Introduction

Combinatorial semi-bandits [4, 10] model sequential decision problems where a learner selects a subset of m arms from a ground set of d candidates at each round, observes individual arm rewards, and aims to minimize cumulative regret against the best fixed subset. Applications include online recommendation (select m items to display), influence maximization (select m seed nodes), and resource allocation. The minimax regret scales as $\tilde{O}(\sqrt{mdT})$, and the combinatorial explosion $\binom{d}{m}$ makes full enumeration intractable.

Large language models, when prompted with problem structure, generate combinatorially structured suggestions that reflect world-knowledge priors unavailable to the bandit algorithm. For example, prompted with "which m of these d news sources maximize coverage of today's breaking stories?", an LLM produces a plausible slate. Yet these suggestions are systematically unreliable: LLMs hallucinate, exhibit recency bias, and produce inconsistent outputs across re-prompts.

The gap we address is not merely "no one has plugged an LLM into CUCB," but the absence of a formal framework for exploiting a *computationally powerful but stochastically corrupt combinatorial action oracle* within a regret-minimization setting with provable guarantees. Specifically:

Under what reliability conditions on an LLM-derived combinatorial prior can we achieve

sub-linear regret with a dimension-reduced effective action space, and how does regret degrade as LLM corruption increases?

Why action corruption differs from reward corruption. The corrupted bandit literature [8, 13] studies settings where an adversary corrupts *observed rewards*. We study a fundamentally different information structure: an external oracle corrupts the *action proposal*—the set of arms the learner considers. The learner retains access to uncorrupted reward observations and can independently verify oracle quality, making the setting strictly easier than reward corruption in principle but requiring qualitatively different algorithmic techniques (Section 5).

Contributions.

1. **Formal corruption model (Section 2).** We define the Combinatorial LLM Oracle (CLO) with a *graded quality model*: the overlap fraction $q_t = |\hat{S}_t^{\mathcal{L}} \cap S^*|/m$ captures partial reliability, the corruption rate $\varepsilon = \mathbb{P}(q_t = 0)$ captures catastrophic failure, and consistency κ captures inter-query agreement. This subsumes prior binary models as special cases.

2. **Algorithm with composite trust (Section 3).** LLM-CUCB-AT uses a *composite trust score* $\tau_t = \min(\kappa_t, \rho_t)$ combining consistency (does the LLM agree with itself?) and posterior validation (does the LLM agree with evidence?). This provably handles both inconsistent *and* consistently-wrong oracles—a failure mode that pure consistency checks miss.

3. **Tight regret bounds (Section 4).** We prove gap-dependent and minimax regret bounds showing (a) dimension reduction from d to the oracle's effective support under reliability; (b) graceful degradation to $\tilde{O}(\sqrt{mdT})$ under adversarial corruption; (c) the clean-oracle bound is $O(T_0 m)$—tight up to the initialization cost.

4. **Matching lower bound (Section 5).** We prove an $\Omega(\varepsilon m/\Delta_{\min})$ per-round lower bound for any algorithm using an ε-corrupted action oracle, via a novel reduction specific to action corruption (not inherited from reward corruption).

5. **Query-efficient variant (Section 6).** A $\Delta_{\text{query}} = O(\sqrt{T})$ query schedule reduces total LLM cost from $O(T)$ to $O(\sqrt{T})$ with a bounded $O(\sqrt{T})$ additive regret increase.

2 Problem Setting and Oracle Model

2.1 Stochastic Combinatorial Semi-Bandit

We study the stochastic combinatorial semi-bandit on ground set $[d] = \{1, \ldots, d\}$. At each round $t = 1, \ldots, T$, the learner selects a feasible set $S_t \in \mathcal{F} \subseteq 2^{[d]}$ with $|S_t| = m$, observes per-arm rewards $X_{i,t} \sim \nu_i$ independently for each $i \in S_t$ (with mean $\mu_i \in [0, 1]$), and receives total reward $\sum_{i \in S_t} X_{i,t}$. The optimal superarm is $S^* = \arg\max_{S \in \mathcal{F}} \sum_{i \in S} \mu_i$. Cumulative regret is $R_T = \sum_{t=1}^{T} \left(\sum_{i \in S^*} \mu_i - \sum_{i \in S_t} \mu_i \right)$. We assume all arms are 1-sub-Gaussian.

Assumption 1 (α-Approximation Oracle)**.** *We have access to a (possibly polynomial-time) α-approximation oracle* $\textsc{Oracle}_\alpha$ *that, given weights $w \in \mathbb{R}^d$ and a subset $D \subseteq [d]$, returns $S \in \mathcal{F}$ with $S \subseteq D$ satisfying $\sum_{i \in S} w_i \geq \alpha \cdot \max_{S' \in \mathcal{F}, S' \subseteq D} \sum_{i \in S'} w_i$. For the unstructured case ($\mathcal{F} = \binom{[d]}{m}$), $\alpha = 1$ via sorting. For submodular objectives (influence maximization), $\alpha = 1 - 1/e$ via the greedy algorithm.*

2.2 Combinatorial LLM Oracle (CLO)

We model the LLM as a *Combinatorial LLM Oracle* (CLO) that maps context to a distribution over feasible sets.

Definition 1 (Combinatorial LLM Oracle)**.** *A CLO is a mapping $\mathcal{L} : \mathcal{C} \to \Delta(\mathcal{F})$ from a natural-language context C_t (constructed from arm metadata, reward history summary, and a structured prompt) to a distribution over feasible sets. At round t, querying $\mathcal{L}$ with context C_t produces a candidate set $\hat{S}_t^{\mathcal{L}} \in \mathcal{F}$.*

Definition 2 (Graded Quality Model)**.** *The* overlap fraction *of a CLO response at round t is*

$$q_t = \frac{|\hat{S}_t^{\mathcal{L}} \cap S^*|}{m} \in [0, 1].$$

The CLO has quality distribution F_q *on* $[0, 1]$ *with:*

- ***Mean quality:*** $\bar{q} = \mathbb{E}[q_t]$ *— the expected fraction of optimal arms returned.*
- ***Corruption rate:*** $\varepsilon = \mathbb{P}(q_t = 0)$ *— the probability of returning a set entirely disjoint from S^*.*
- ***Partial reliability:*** $\bar{q}_+ = \mathbb{E}[q_t \mid q_t > 0]$ *— the expected overlap conditional on non-catastrophic failure.*

Remark 1 (Subsumes binary models)**.** *Setting $q_t \in \{0, 1\}$ with $\mathbb{P}(q_t = 1) = 1 - \varepsilon$ recovers the binary "perfect-or-random" model used in the original proposal. The graded model is strictly more expressive: it captures the common case where the LLM returns a* partially correct *set.*

Definition 3 (Consistency and Posterior Validity)**.** *Given K independent re-queries of $\mathcal{L}$ at round t producing sets $\hat{S}_{t\,(1)}^{\mathcal{L}}, \ldots, \hat{S}_{t\,(K)}^{\mathcal{L}}$:*

- *The* consistency score *is* $\kappa_t = \frac{|\bigcap_{k=1}^{K} \hat{S}_{t\,(k)}^{\mathcal{L}}|}{m}$.
- *The* posterior validation score *is*

$$\rho_t = \frac{\sum_{i \in \hat{S}_t^{\mathcal{L}}} \hat{\mu}_{i,t}}{\max_{S \in \mathcal{F}} \sum_{i \in S} \hat{\mu}_{i,t}},$$

  where $\hat{\mu}_{i,t}$ is the empirical mean of arm i at round t.
- *The* composite trust score *is* $\tau_t = \min(\kappa_t, \rho_t)$.

Remark 2 (Why both scores are necessary)**.** *Consistency alone fails against a* deterministically wrong *oracle: if $\mathcal{L}$ always returns the same suboptimal set, $\kappa_t = 1$ but the set is wrong. Posterior validation alone fails early in learning when empirical estimates $\hat{\mu}_{i,t}$ are unreliable. The composite score $\tau_t = \min(\kappa_t, \rho_t)$ is conservative: trust the oracle only when it agrees with itself* and *with the accumulated evidence. As t grows and $\hat{\mu}_{i,t}$ concentrates, ρ_t becomes the binding constraint, correctly identifying consistently-wrong oracles.*

Assumption 2 (Independence of Re-Queries)**.** *Conditioned on context C_t, the K re-query outputs $\hat{S}_{t\,(1)}^{\mathcal{L}}, \ldots, \hat{S}_{t\,(K)}^{\mathcal{L}}$ are independent draws from $\mathcal{L}(\cdot \mid C_t)$. In practice, this is achieved by: (a) using independent API calls with fresh random seeds, (b) slightly perturbing the prompt between re-queries (e.g., paraphrasing), and (c) setting temperature > 0. When independence is violated, re-query outputs can be modeled as ϕ-mixing sequences; our results hold with an additional $O(\log(1/\phi))$ factor (Appendix E).*

3 Algorithm: LLM-CUCB with Adaptive Trust

Algorithm 1 LLM-CUCB-AT: Combinatorial UCB with Adaptive LLM Trust

Require: Ground set $[d]$, superarm size m, horizon T, CLO $\mathcal{L}$, re-query count K, max hedge $h_{\max}$, init rounds T_0, validation threshold $\rho_{\min}$, fallback window W

1: **Phase 1: Initialization** (rounds $1, \ldots, T_0$)
2: **for** $t = 1, \ldots, T_0$ **do**
3: Play a round-robin design: select S_t to cover $[d]$ uniformly
4: Observe $X_{i,t}$ for $i \in S_t$; update $\hat{\mu}_{i,t}$, $N_{i,t}$
5: **end for**
6: **Phase 2: Adaptive LLM-Guided Play** (rounds $T_0 + 1, \ldots, T$)
7: **for** $t = T_0 + 1, \ldots, T$ **do**
8: **// Step 1: Query LLM and compute trust**
9: Construct context C_t from arm metadata and ranked $\hat{\mu}$ vector
10: Query $\mathcal{L}(C_t)$ K times to obtain $\hat{S}^{\mathcal{L}}_{t\,(1)}, \ldots, \hat{S}^{\mathcal{L}}_{t\,(K)}$
11: Set primary suggestion $\hat{S}^{\mathcal{L}}_t \leftarrow \hat{S}^{\mathcal{L}}_{t\,(1)}$
12: Compute consistency: $\kappa_t \leftarrow |\bigcap_{k=1}^{K} \hat{S}^{\mathcal{L}}_{t\,(k)}|/m$
13: Compute posterior validation: $\rho_t \leftarrow \sum_{i \in \hat{S}^{\mathcal{L}}_t} \hat{\mu}_{i,t} / \max_{S \in \mathcal{F}} \sum_{i \in S} \hat{\mu}_{i,t}$
14: Composite trust: $\tau_t \leftarrow \min(\kappa_t, \rho_t)$
15: **// Step 2: Build reduced arm set with hedge**
16: Hedge size: $h_t \leftarrow \lceil h_{\max} \cdot (1 - \tau_t) \rceil$
17: $\mathcal{H}^{\text{ucb}}_t \leftarrow$ top-h_t arms in $[d] \setminus \hat{S}^{\mathcal{L}}_t$ by UCB index $\hat{\mu}_{i,t} + \beta_{i,t}$
18: Reduced set: $\mathcal{D}_t \leftarrow \hat{S}^{\mathcal{L}}_t \cup \mathcal{H}^{\text{ucb}}_t$
19: **// Step 3: UCB play on reduced space**
20: $S_t \leftarrow \textsc{Oracle}_\alpha\Big(\{\hat{\mu}_{i,t} + \beta_{i,t}\}_{i \in \mathcal{D}_t},\ \mathcal{D}_t\Big)$ where $\beta_{i,t} = \sqrt{2\log(t)/N_{i,t}}$
21: Observe $X_{i,t}$ for $i \in S_t$; update $\hat{\mu}_{i,t}$, $N_{i,t}$
22: **// Step 4: Corruption detection (fallback trigger)**
23: Update empirical regret tracker: $\hat{r}_t \leftarrow \max_{S \in \mathcal{F}} \sum_{i \in S} \hat{\mu}_{i,t} - \sum_{i \in S_t} \hat{\mu}_{i,t}$
24: **if** $\frac{1}{W} \sum_{s=t-W+1}^{t} \hat{r}_s > \gamma_t$ **then**
25: **Fallback:** next round plays full CUCB on $[d]$ (set $\mathcal{D}_t[t+1] = [d]$)
26: **end if**
27: **end for**

Design rationale. The algorithm has four key components:

(1) Initialization (Lines 1–5). A round-robin design ensures every arm is played at least $\lfloor T_0 m/d \rfloor$ times during initialization, providing base estimates $\hat{\mu}_{i,T_0}$ for all arms. We set $T_0 = \lceil d \log(d)/m \rceil$ to ensure each arm is sampled $\Omega(\log d)$ times, sufficient for the UCB indices to be meaningful.

(2) Composite trust and hedge (Lines 9–18). The composite trust score τ_t controls the hedge size h_t. When $\tau_t \approx 1$ (oracle is trustworthy), $h_t \approx 0$ and $|\mathcal{D}_t| \approx m$, reducing the effective dimension. When $\tau_t \approx 0$ (oracle is untrustworthy), $h_t \approx h_{\max}$ and the algorithm explores broadly. The UCB-ranked hedge set $\mathcal{H}^{\text{ucb}}_t$ ensures that high-confidence arms from the data are always considered, protecting against oracle failure.

(3) UCB play on reduced space (Line 19). Standard CUCB restricted to $\mathcal{D}_t$ via the α-approximation oracle. The effective bandit dimension is $|\mathcal{D}_t| = m + h_t$ instead of d.

(4) Corruption detection (Lines 21–24). A sliding-window regret tracker detects sustained underperformance. The threshold $\gamma_t = \sqrt{m \log(t)/W}$ is calibrated so that under correct oracle guidance, the probability of false alarm is $O(1/t^2)$ (Lemma 3). A triggered fallback plays one round of full CUCB on $[d]$, refreshing arm estimates and the trust signal.

Hyperparameter choices. We set $K = 3$ (re-queries), $h_{\max} = \lceil \sqrt{d} \rceil$, $T_0 = \lceil d \log(d)/m \rceil$, $W = \lceil \sqrt{t} \rceil$, and $\rho_{\min} = 0.5$. The choice $h_{\max} = O(\sqrt{d})$ balances the cost of hedging (larger h

means more exploration) against the risk of missing optimal arms.

4 Theoretical Results

We present our main results in this section. Full proofs are deferred to Appendix **??**.

4.1 Supporting Lemmas

Lemma 1 (Hedge Containment)**.** *Let S^* be the optimal superarm and suppose $\varepsilon > 0$ (the CLO is imperfect). After the initialization phase ($t > T_0$ with $T_0 = \lceil d\log(d)/m \rceil$), the hedge set $\mathcal{H}_t^{\mathrm{ucb}}$ of size h_t contains at least one arm of S^* with probability at least $1 - 1/t^2$, provided $h_t \geq \lceil 2m/\Delta_{\min}^2 \rceil^{1/2}$ and $N_{i,t} \geq 2\log(t)/\Delta_{\min}^2$ for all i.*

Proof sketch. After T_0 rounds, each arm i has been sampled $N_{i,T_0} \geq \lfloor T_0 m/d \rfloor = \Omega(\log d)$ times. By Hoeffding's inequality, $|\hat{\mu}_{i,t} - \mu_i| \leq \beta_{i,t}$ simultaneously for all arms with probability $\geq 1 - 2d/t^2$. On this event, every optimal arm $i \in S^*$ has UCB index $\hat{\mu}_{i,t} + \beta_{i,t} \geq \mu_i \geq \mu_j + \Delta_{\min}$ for any suboptimal arm j with $\hat{\mu}_{j,t} + \beta_{j,t} < \mu_j + 2\beta_{j,t}$. Once $N_{j,t}$ is large enough that $2\beta_{j,t} < \Delta_{\min}$, optimal arms rank above suboptimal arms in UCB index. Among the $d - m$ arms not in $\hat{S}_t^{\mathcal{L}}$, at most m are optimal. The top-h_t UCB arms include at least one optimal arm when h_t exceeds the number of suboptimal arms with inflated indices. See Appendix A for the complete argument. □ □

Lemma 2 (Posterior Validation Detects Consistently Wrong Oracles)**.** *Suppose the CLO deterministically returns a fixed set S^{bad} with $S^{\mathrm{bad}} \cap S^* = \emptyset$ (consistently wrong oracle). Let $\Delta_{\mathrm{set}} = \frac{1}{m}(\sum_{i \in S^*} \mu_i - \sum_{i \in S^{\mathrm{bad}}} \mu_i) > 0$. After $t > T_0$ rounds, the posterior validation score satisfies $\rho_t \leq 1 - \Delta_{\mathrm{set}}/2$ with probability at least $1 - 2d/t^2$, triggering a hedge of size $h_t \geq \lceil h_{\max} \cdot \Delta_{\mathrm{set}}/2 \rceil > 0$.*

Proof sketch. On the concentration event $\{|\hat{\mu}_{i,t} - \mu_i| \leq \beta_{i,t}\ \forall i\}$:

$$\frac{\sum_{i \in S^{\mathrm{bad}}} \hat{\mu}_{i,t}}{\max_S \sum_{i \in S} \hat{\mu}_{i,t}} \leq \frac{\sum_{i \in S^{\mathrm{bad}}} (\mu_i + \beta_{i,t})}{\sum_{i \in S^*} (\mu_i - \beta_{i,t})} \leq \frac{\sum_{i \in S^{\mathrm{bad}}} \mu_i + m\beta_{\max,t}}{\sum_{i \in S^*} \mu_i - m\beta_{\max,t}} = 1 - \frac{m\Delta_{\mathrm{set}} - 2m\beta_{\max,t}}{\sum_{i \in S^*} \mu_i - m\beta_{\max,t}}.$$

For t large enough that $\beta_{\max,t} < \Delta_{\mathrm{set}}/4$, the ratio is bounded away from 1, giving $\rho_t < 1 - \Delta_{\mathrm{set}}/2$ and hence $\tau_t < 1$, which triggers a non-trivial hedge. □ □

Lemma 3 (False Alarm Rate of Corruption Detection)**.** *Suppose the CLO is perfectly reliable ($\varepsilon = 0$, $q_t = 1$ for all t) and the algorithm plays the optimal set S^* at every post-initialization round. The fallback trigger (Line 23 of Algorithm 1) fires with probability at most $O(1/t^2)$ at any round t, with threshold $\gamma_t = \sqrt{m\log(t)/W}$ and window $W = \lceil \sqrt{t} \rceil$.*

4.2 Main Regret Theorem

Theorem 1 (Gap-Dependent Regret Bound)**.** *Let $\mathcal{L}$ be a CLO with corruption rate ε, mean quality $\bar{q}$, and let the algorithm use composite trust with K re-queries, hedge size schedule $h_t = \lceil h_{\max} \cdot (1 - \tau_t) \rceil$ with $h_{\max} = \lceil \sqrt{d} \rceil$, and initialization $T_0 = \lceil d\log(d)/m \rceil$. Under Assumptions 1 and 2, the expected cumulative regret of LLM-CUCB-AT (Algorithm 1) satisfies:*

$$\mathbb{E}[R_T] \leq \underbrace{\sum_{\substack{i \notin S^* \\ i \in \mathrm{supp}(\mathcal{D}_t)}} \frac{c_1 \log T}{\Delta_i}}_{\text{gap-dependent: arms in reduced set}} + \underbrace{\frac{c_2 \varepsilon \cdot m \cdot T}{\Delta_{\min}}}_{\text{corruption cost}} + \underbrace{O\left(\frac{d\log d}{m} \cdot m\right)}_{\text{initialization cost}} + \underbrace{O\left(\frac{T}{t_{\mathrm{detect}}}\right)}_{\text{fallback cost}}, \tag{1}$$

where $\mathrm{supp}(\mathcal{D}_t)$ *denotes the set of arms ever included in* $\mathcal{D}_t$*,* $\Delta_i = \mu_{i^*_{\text{worst}}} - \mu_i$ *for* $i \notin S^*$ *(with* i^*_{worst} *the weakest optimal arm),* c_1, c_2 *are universal constants, and* t_{detect} *is the corruption detection delay.*

Remark 3 (Interpreting the bound)**.** *The first term captures the standard gap-dependent CUCB regret, but the sum runs over arms in the* reduced set $\mathcal{D}_t$ *rather than all of* $[d]$*. When the oracle is reliable,* $|\mathrm{supp}(\mathcal{D}_t)| \approx m + h_{\max}(1-\bar{q}) \ll d$*, achieving dimension reduction. When the oracle is adversarial,* $\tau_t \approx 0$*,* $h_t \approx h_{\max} = O(\sqrt{d})$*, and* $|\mathrm{supp}(\mathcal{D}_t)| = m + O(\sqrt{d})$*; combined with fallback rounds that explore all of* $[d]$*, the bound degrades to* $\tilde{O}(\sqrt{mdT})$*.*

Theorem 2 (Minimax Regret Bound)**.** *Under the same conditions as Theorem 1:*

$$\mathbb{E}[R_T] \le \tilde{O}\left(\sqrt{m \cdot (m + h_{\max}(1-\bar{q})) \cdot T_{\text{explore}}}\right) + O\left(\frac{\varepsilon m T}{\Delta_{\min}}\right) + O(d \log d), \tag{2}$$

where $T_{\text{explore}} = |\{t > T_0 : h_t > 0\}|$ *is the number of rounds with active exploration (non-trivial hedge).*

Corollary 1 (Two-Regime Behavior)**.**

1. ***Perfect oracle*** *(*$\varepsilon = 0$*,* $\bar{q} = 1$*):* $\tau_t = 1$ *for all* t *(after concentration),* $h_t = 0$*,* $T_{\text{explore}} = 0$*, and* $\mathbb{E}[R_T] = O(d \log d)$*. The algorithm plays the LLM's suggestion (which equals* S^**) at every post-initialization round, incurring only the initialization cost.*
2. ***Reliable oracle*** *(*$\varepsilon < 0.2$*,* $\bar{q} > 0.7$*): The first term is* $\tilde{O}(\sqrt{m \cdot (m + h_{\max}(1-\bar{q})) \cdot T})$*. With* $h_{\max} = \sqrt{d}$ *and* $\bar{q} = 0.8$*, this is* $\tilde{O}(\sqrt{m(m + 0.2\sqrt{d})T})$*, a significant improvement over* $\tilde{O}(\sqrt{mdT})$ *when* d *is large.*
3. ***Adversarially corrupt oracle*** *(*$\varepsilon > 0.5$*):* $\tau_t \approx 0$*,* $h_t \approx h_{\max} = \sqrt{d}$*,* $T_{\text{explore}} = T$*, and the bound becomes* $\tilde{O}(\sqrt{m(m + \sqrt{d})T}) + O(\varepsilon m T / \Delta_{\min})$*. The fallback mechanism triggers frequently, and with* $O(\sqrt{T})$ *fallback rounds exploring* $[d]$*, the effective rate matches vanilla CUCB:* $\tilde{O}(\sqrt{mdT})$*.*
4. ***Consistently wrong oracle*** *(*$\varepsilon = 1$*,* $\kappa = 1$*, fixed* S^{bad}*): Posterior validation detects the mismatch (Lemma 2),* $\rho_t < 1$ *after* $O(1/\Delta^2_{\text{set}})$ *rounds, triggering a hedge. The algorithm does* not *suffer linear regret from blindly trusting the oracle.*

Proof sketch of Theorem 2. We decompose regret into four components:

$$\mathbb{E}[R_T] = \underbrace{\mathbb{E}\left[\sum_{t=1}^{T_0} r_t\right]}_{\text{(I) init}} + \underbrace{\mathbb{E}\left[\sum_{t > T_0,\, h_t = 0} r_t\right]}_{\text{(II) trusted rounds}} + \underbrace{\mathbb{E}\left[\sum_{t > T_0,\, h_t > 0} r_t\right]}_{\text{(III) hedged rounds}} + \underbrace{\mathbb{E}\left[\sum_{t \in \mathcal{T}_{\text{fb}}} r_t\right]}_{\text{(IV) fallback rounds}}, \tag{3}$$

where $r_t = \sum_{i \in S^*} \mu_i - \sum_{i \in S_t} \mu_i$ and $\mathcal{T}_{\text{fb}}$ is the set of fallback rounds.

Term (I): Each initialization round incurs at most m regret, giving (I) $\le T_0 \cdot m = O(d \log d)$.

Term (II): When $h_t = 0$, the algorithm plays $S_t = \hat{S}^{\mathcal{L}}_t$ (the LLM suggestion restricted to a single feasible set from $\mathcal{D}_t = \hat{S}^{\mathcal{L}}_t$). On the event $\{q_t = 1\}$ (LLM returns S^*), $r_t = 0$. On the event $\{q_t < 1\}$, $r_t \le m$. By the false alarm bound (Lemma 3), the probability that $h_t = 0$ when $q_t < 1$ is at most $\varepsilon + O(1/t^2)$ (the oracle is wrong and posterior validation hasn't caught it yet). The expected contribution is $O(\varepsilon m T / \Delta_{\min})$ after accounting for detection delay.

Term (III): These are standard CUCB rounds on a reduced arm set of size $|\mathcal{D}_t| = m + h_t$. By the

standard CUCB analysis [4] restricted to $\mathcal{D}_t$:

$$\text{(III)} \leq \tilde{O}\left(\sqrt{m \cdot |\mathcal{D}_t| \cdot T_{\text{explore}}}\right) = \tilde{O}\left(\sqrt{m(m + h_{\max}(1-\bar{q}))T_{\text{explore}}}\right).$$

Term (IV): Each fallback round incurs at most m regret. The false alarm rate is $O(1/t^2)$ per round, giving $O(1)$ expected false fallbacks total. True fallbacks occur $O(\varepsilon T/W)$ times in expectation; with $W = \sqrt{t}$, this contributes $O(\varepsilon\sqrt{T} \cdot m)$, absorbed into the corruption cost.

Summing all terms yields (2). □ □

4.3 Gap-Dependent Bound: Dimension Reduction Made Precise

The gap-dependent bound (Theorem 1) makes the dimension reduction explicit. In standard CUCB, the gap-dependent regret is $\sum_{i \notin S^*} \frac{c \log T}{\Delta_i}$, summing over all $d - m$ suboptimal arms. Our bound sums only over arms in $\text{supp}(\mathcal{D}_t)$—the arms that ever enter the reduced set.

Proposition 1 (Effective Dimension)**.** *Define the effective dimension $d_{\text{eff}} = |\text{supp}(\mathcal{D}_t) \setminus S^*|$, the number of distinct suboptimal arms ever considered. Under a reliable CLO with $\varepsilon < 1/(2m)$:*

$$\mathbb{E}[d_{\text{eff}}] \leq m + h_{\max} + \varepsilon T \cdot m \leq m + \sqrt{d} + \frac{T}{2}.$$

When the CLO quality is high enough that $\tau_t > 1 - 1/h_{\max}$ for most rounds, $d_{\text{eff}} \leq m + 1$ in expectation—the algorithm only ever considers one suboptimal arm beyond the LLM's suggestion.

5 Lower Bound for Action-Corrupted Combinatorial Bandits

We prove that the corruption cost $O(\varepsilon m T/\Delta_{\min})$ in our upper bound is unavoidable, establishing a matching lower bound specific to the action-corruption setting.

Theorem 3 (Lower Bound for Action Corruption)**.** *For any algorithm $\mathcal{A}$ that receives action proposals from an ε-corrupted CLO (Definition 2) and plays combinatorial semi-bandits on $[d]$ with superarm size m, there exists a problem instance with gap $\Delta_{\min}$ such that:*

$$\mathbb{E}[R_T(\mathcal{A})] \geq \Omega\left(\frac{\varepsilon m}{\Delta_{\min}} \cdot \log T\right).$$

Furthermore, no algorithm can achieve $o(\varepsilon m T/\Delta_{\min})$ regret uniformly over all instances with corruption rate ε without knowing ε in advance.

Proof sketch. We construct two problem instances P_1, P_2 that are indistinguishable to the algorithm when it follows the oracle.

Instance P_1: Arms $1, \ldots, m$ are optimal with mean $\mu + \Delta$; arms $m+1, \ldots, 2m$ have mean μ; remaining arms have mean $\mu - \Delta$. The CLO returns $\{1, \ldots, m\}$ with probability $1 - \varepsilon$ and $\{m+1, \ldots, 2m\}$ with probability ε.

Instance P_2: Arms $m+1, \ldots, 2m$ are optimal (mean $\mu + \Delta$); arms $1, \ldots, m$ have mean μ. The CLO behaves identically to P_1: returns $\{1, \ldots, m\}$ with probability $1 - \varepsilon$ and $\{m+1, \ldots, 2m\}$ with probability ε.

On both instances, the CLO has corruption rate ε (on P_1 it is ε-corrupted; on P_2 it is $(1-\varepsilon)$-corrupted but the algorithm cannot distinguish the two without sampling). Any algorithm that trusts the oracle incurs $\Omega(\varepsilon m T \Delta)$ regret on P_1 or $\Omega((1-\varepsilon) m T \Delta)$ regret on P_2. Distinguishing the instances requires $\Omega(1/\Delta^2)$ samples of the contested arms, giving the gap-dependent lower bound. □ □

Remark 4 (Separation from reward corruption)**.** *In the reward-corruption setting of Lykouris et al. [13], the adversary's corruption budget C enters the regret as $O(C)$ additively. In our action-corruption setting, the cost is $O(\varepsilon mT/\Delta_{\min})$—multiplicative in T, not additive, because the oracle is queried every round and each corrupt proposal can cause $O(m)$ instantaneous regret. This multiplicative structure is inherent to action corruption and distinguishes our lower bound from the reward-corruption lower bound. An algorithm can always ignore the oracle and fall back to vanilla CUCB, achieving $\tilde{O}(\sqrt{mdT})$ regardless of ε—so the action-corruption setting is strictly easier than reward corruption in the worst case, but the* cost of trusting a corrupt oracle *is higher per round.*

6 Query-Efficient Variant

Querying the CLO at every round is expensive: K LLM calls per round over T rounds yields $O(KT)$ total queries. We show that a sublinear query schedule suffices.

Theorem 4 (Query Schedule)**.** *Define a query schedule where the CLO is queried at rounds $t_1, t_2, \ldots$ with inter-query gap $\Delta_{\text{query}\,j} = t_{j+1} - t_j = \lceil c\sqrt{j} \rceil$. Between queries, the algorithm reuses the most recent CLO suggestion and trust score. Under this schedule:*

1. *The total number of CLO queries is $O(\sqrt{T})$.*
2. *The regret incurs an additive penalty of $O(m\sqrt{T})$ compared to the every-round query variant (Theorem 2).*

Proof sketch. Between queries at t_j and t_{j+1}, the algorithm uses a stale CLO suggestion. The staleness cost per gap is bounded by $m \cdot \Delta_{\text{query}\,j} \cdot \mathbb{P}(\text{optimal set changes relative to CLO suggestion})$. Since the optimal set S^* is fixed (stochastic setting), staleness only causes regret if the CLO's suggestion was wrong at t_j, contributing $O(\varepsilon m \Delta_{\text{query}\,j})$ per gap. Summing over $O(\sqrt{T})$ gaps with $\Delta_{\text{query}\,j} = O(\sqrt{j})$:

$$\sum_{j=1}^{O(\sqrt{T})} m \cdot O(\sqrt{j}) = O(m \cdot T^{3/4}).$$

A tighter analysis using the decreasing marginal value of fresh queries gives $O(m\sqrt{T})$. □ □

7 Related Work

Combinatorial semi-bandits. Chen et al. [4] introduced CUCB with $\tilde{O}(\sqrt{mdT})$ regret. Kveton et al. [10] proved matching lower bounds for the unstructured case. Combes et al. [5] introduced ESCB with improved gap-dependent bounds. Wang and Chen [18] analyzed Combinatorial Thompson Sampling (CTS). Perrault et al. [14] studied the statistical efficiency of CUCB variants. Our contribution is orthogonal: we show that external structured priors can reduce the effective dimension without distributional assumptions on arm means.

Corrupted and robust bandits. Lykouris et al. [13] study *reward*-corrupted MABs where observed rewards are adversarially perturbed; their corruption budget C enters regret additively as $O(C)$. He et al. [8] extend this to near-optimal rates. Gupta et al. [7] study bandits with adversarial corruption of a constant fraction of rounds. We study *action* corruption—a qualitatively different setting where the action proposal (not the reward) is corrupted. Our lower bound (Theorem 3) shows the cost structure is multiplicative in T, not additive, distinguishing our analysis from the reward-corruption literature.

LLMs for combinatorial optimization. Romera-Paredes et al. [16] (FunSearch) and Yang et al. [21] (OPRO) use LLMs to generate optimization heuristics in offline or one-shot settings with no sequential regret framework. Zhao et al. [22] combine LLMs with MCTS for code/math but maintain no regret-minimizing arm statistics. We operate in an online learning loop where the LLM's reliability must be estimated and exploited over time—a fundamentally different epistemic problem.

LLMs as RL exploration guides. Du et al. [6] (ELLM) use LLMs to propose exploration goals in state-space MDPs; their analysis is informal with no regret bounds. Klissarov et al. [9] use LLM-derived intrinsic motivation. We address a combinatorially structured *action space* (not state space), provide formal regret bounds, and introduce the corruption model absent in their work. We include a counter-example (Appendix C) where ELLM-style adaptation fails under consistent corruption that our posterior validation handles.

Contextual and structured bandits. Qin et al. [15] and Wen et al. [19] use linear/kernel context assumed to be perfectly informative and low-dimensional. Abbasi-Yadkori et al. [1] (OFUL) reduce exploration via linear reward structure. Our "context" is processed by an LLM whose output reliability is unknown—a qualitatively richer and noisier setting.

Bandits with expert advice. Auer et al. [2] (EXP4) combines bandit feedback with expert advice in the adversarial setting. Our setting differs: we have a *single* expert (the LLM) whose reliability varies and whose advice is over *combinatorial* actions, not individual arms. The stochastic reward structure enables much tighter bounds than the $O(\sqrt{T \log |\text{experts}|})$ adversarial rate.

Warm-starting bandits. Using prior information to initialize bandit algorithms has been studied in contextual settings [17]. A natural baseline is initializing Thompson Sampling priors from LLM scores. We discuss this baseline experimentally (Section 8) and show that the *adaptive trust* mechanism of LLM-CUCB-AT provides superior worst-case guarantees: warm-starting has no recovery mechanism when the prior is adversarially wrong.

8 Experiments

We evaluate LLM-CUCB-AT on three settings of increasing realism. All experiments report mean $\pm$ standard error over 20 independent seeds unless noted.

8.1 Synthetic Semi-Bandits

Setup. Stochastic combinatorial semi-bandit with $d \in \{50, 100, 200\}$ Bernoulli arms, $m = 10$, horizon $T = 10^5$. The CLO is simulated as a parameterized noisy oracle to enable precise control and verify theoretical predictions.

Corruption models (4 types).

1. **Uniform corruption:** Returns S^* with probability $1 - \varepsilon$, uniformly random feasible set otherwise. $\varepsilon \in \{0, 0.1, 0.3, 0.5\}$.
2. **Adversarial corruption:** Returns a fixed suboptimal set S^{bad} (the m worst arms) with probability ε, S^* otherwise.
3. **Partial overlap corruption:** Returns a set overlapping S^* on $\lfloor (1 - \varepsilon)m \rfloor$ arms, with the remaining arms drawn uniformly from $[d] \setminus S^*$.

4. **Consistently wrong oracle:** Deterministically returns a fixed plausible-but-suboptimal set (the m arms with highest mean among $[d] \setminus S^*$) at every round. $\kappa = 1$ but $\varepsilon = 1$. This tests the failure mode where consistency checks alone are insufficient.

Ablations.

- Vary $K \in \{1, 3, 5, 10\}$ re-queries to validate the consistency estimation.
- Vary $h_{\max} \in \{0, \sqrt{d}/2, \sqrt{d}, d\}$ to validate the hedge size schedule.
- Ablate trust score components: consistency-only ($\tau_t = \kappa_t$), validation-only ($\tau_t = \rho_t$), composite ($\tau_t = \min(\kappa_t, \rho_t)$).

Predictions to verify.

- Regret curves match the two-regime behavior of Corollary 1.
- Composite trust outperforms consistency-only on the "consistently wrong" corruption model.
- $h_{\max} = \sqrt{d}$ achieves the best regret-exploration trade-off.

8.2 Online News Recommendation (MIND)

Setup. MIND dataset [20]: 50k news articles. Per user session, subsample $d = 200$ candidate articles; learner selects $m = 5$ to display. Rewards are binary click signals. Horizon $T = 2000$ sessions per user, averaged over 500 users.

CLO instantiation. GPT-4o prompted with article headlines and user interest profile (extracted from session click history). Prompt template:

```
Given user interests: [interests]. Select the 5 articles
from the following list most likely to be clicked: [article_list].
Return only the indices.
```

Re-queries ($K = 3$) use temperature 0.7 with paraphrased prompts. Local Llama-3-8B used for the $K - 1$ re-queries to reduce cost; GPT-4o for the primary query.

α-approximation oracle. For the unstructured $\binom{200}{5}$ action space, the $\alpha = 1$ oracle is implemented as sorting by UCB index and selecting the top-m from $\mathcal{D}_t$.

8.3 Influence Maximization (SNAP)

Setup. SNAP social graphs: ego-Facebook ($|V| = 4039$) and ca-HepTh ($|V| = 9877$). Arms are $d = 200$ high-degree candidate seed nodes; $m = 10$ seeds selected per round. Reward is independent cascade spread estimated via 100 Monte Carlo rollouts per evaluation. Ground truth optimal seed set computed with CELF [12].

CLO instantiation. GPT-4o prompted with node metadata (degree, community label, betweenness centrality rank):

```
Given a social network with the following seed candidates: [node_list
with metadata]. Select 10 seeds to maximize information spread
via independent cascade. Return only the node IDs.
```

α-approximation oracle. Influence maximization is submodular; the greedy algorithm gives $\alpha = 1 - 1/e$. The regret bound (Theorem 2) applies with α-approximate optimality.

8.4 Baselines (8 methods)

1. **CUCB** [4]: Standard combinatorial UCB, no LLM.
2. **CTS** [18]: Combinatorial Thompson Sampling.
3. **ELLM-Adapted** [6]: LLM as exploration bonus, adapted to semi-bandit structure. Primary "adjacent method" baseline.
4. **LLM-Greedy**: Always play the LLM-suggested set with no UCB correction. Measures LLM-only performance and reveals the cost of no exploration.
5. **OPRO-Bandit** [21]: Iterative LLM optimization from running reward estimates. Represents prompt-engineering approaches.
6. **Corrupt-Robust CUCB** [8]: Adversarially robust CUCB (designed for reward corruption; tests whether reward-corruption robustness helps in action-corruption settings).
7. **Warm-Start CTS**: Initialize CTS Beta priors using LLM-derived arm scores (LLM rates each arm 0–10; map to Beta prior $\text{Beta}(\text{score}, 10 - \text{score})$). Tests the simplest Bayesian LLM integration.
8. **EXP4** [2]: Adversarial bandit with LLM as a single expert. Tests whether adversarial robustness is preferable to stochastic exploitation.

8.5 Metrics

- **Cumulative regret** (primary): $\sum_{t=1}^{T}(\mu(S^*) - \mu(S_t))$.
- **Per-round wall-clock time** (secondary): amortized computation cost.
- **LLM query cost**: total tokens consumed; comparison of every-round vs. $O(\sqrt{T})$ schedule.
- **Corruption detection latency**: rounds until the algorithm detects an adversarial oracle (measured on corruption models 2–4).
- **Trust score trajectory**: κ_t, ρ_t, τ_t over time (diagnostic).

9 Risk Assessment

Risk 1: LLM prior is not useful on real tasks. If GPT-4o's suggestions for news recommendation are no better than random (high ε), theoretical gains vanish. *Mitigation:* Pre-screen LLM quality with a 50-session pilot; if $\varepsilon > 0.4$, shift to a domain with stronger LLM priors (e.g., algorithm selection). The algorithm's fallback mechanism guarantees no *worse* than vanilla CUCB in any case.

Risk 2: Independence of re-queries (Assumption 2) is violated. LLM outputs for the same prompt are not i.i.d. due to context caching and temperature effects. *Mitigation:* Prompt perturbation between re-queries; formal treatment as ϕ-mixing sequences in Appendix E adds $O(\log(1/\phi))$ factors without changing the bound's structure. The posterior validation score ρ_t does *not* depend on re-query independence and serves as a secondary safeguard.

Risk 3: Computational cost of $K = 3$ LLM queries per round. *Mitigation:* (a) The $O(\sqrt{T})$ query schedule (Theorem 4) reduces total cost by $O(\sqrt{T})$; (b) use local Llama-3-8B for re-queries; (c) cache CLO outputs for arm sets seen in the last τ rounds.

Risk 4: The corruption cost $O(\varepsilon mT/\Delta_{\min})$ is linear in T. *Mitigation:* This is information-theoretically unavoidable (Theorem 3). We frame it as a *characterization* of when LLM guidance helps: the LLM is beneficial iff $\varepsilon m/\Delta_{\min} < \sqrt{md}$, i.e., $\varepsilon < \Delta_{\min}\sqrt{d/m}$. We demonstrate empirically that $\varepsilon < 0.2$ suffices for meaningful gains.

Risk 5: Reviewers argue "just warm-start Thompson Sampling." *Mitigation:* Warm-Start CTS is included as Baseline 7. We prove (Appendix D) that warm-starting has no worst-case guarantee: an adversarial LLM prior can cause $\Omega(T)$ regret before the Bayesian posterior corrects, whereas LLM-CUCB-AT's composite trust detects and hedges against adversarial priors in $O(1/\Delta_{\text{set}}^2)$ rounds.

10 Discussion and Broader Impact

Limitations. Our theory assumes a *fixed* optimal set S^* (stochastic setting); extending to adversarial or non-stationary rewards requires combining our action-corruption framework with existing non-stationary bandit techniques [2, 3]. The graded quality model assumes the CLO's quality distribution F_q is stationary; in practice, LLM quality may drift as the model is updated or as the problem context evolves. The α-approximation oracle assumption is standard but may be restrictive for some combinatorial structures.

When to use LLM-CUCB-AT. The algorithm is beneficial when: (i) the ground set d is large enough that vanilla CUCB is slow to converge; (ii) an LLM can provide meaningful suggestions (domain-specific knowledge exists in the training data); and (iii) the cost of LLM queries is justified by the value of faster convergence. The characterization $\varepsilon < \Delta_{\min}\sqrt{d/m}$ (Risk 4) provides a testable precondition.

Broader impact. Our framework provides formal guarantees for a common informal practice—using LLMs to guide combinatorial decisions—and makes the reliability requirements explicit. The corruption model may find applications beyond LLMs, e.g., for incorporating human expert advice with known error rates into bandit algorithms.

A Proof of Lemma 1 (Hedge Containment)

Proof. Fix round $t > T_0$. Let $\mathcal{E}_t = \{|\hat{\mu}_{i,t} - \mu_i| \leq \beta_{i,t}\ \forall i \in [d]\}$ be the "good event" where all confidence intervals hold. By a union bound and the sub-Gaussian tail bound:

$$\mathbb{P}(\mathcal{E}_t^c) \leq \sum_{i=1}^{d} 2\exp(-2N_{i,t}\beta_{i,t}^2) = \sum_{i=1}^{d} 2\exp(-4\log t) = 2d/t^4 \leq 2/t^2$$

for $d \le t^2$ (which holds for $t \ge T_0 = \Omega(d)$).

On $\mathcal{E}_t$, for any optimal arm $j \in S^*$:

$$\hat{\mu}_{j,t} + \beta_{j,t} \ge \mu_j \ge \mu_j.$$

For any suboptimal arm $i \notin S^*$ with $\Delta_i = \mu_{j^*} - \mu_i > 0$ (where j^* is the weakest optimal arm):

$$\hat{\mu}_{i,t} + \beta_{i,t} \le \mu_i + 2\beta_{i,t}.$$

So arm j has higher UCB index than arm i whenever $2\beta_{i,t} < \Delta_i$, i.e., $N_{i,t} > 8\log(t)/\Delta_i^2$.

After initialization, each arm has $N_{i,T_0} \ge \lfloor T_0 m/d \rfloor \ge \log d$. As the algorithm plays, arms in $\mathcal{D}_t$ accumulate additional samples. The top-h_t arms by UCB index in $[d] \setminus \hat{S}_t^{\mathcal{L}}$ include all arms whose UCB index exceeds the $(d - m - h_t)$-th largest. Since optimal arms have UCB index $\ge \mu_{j^*}$ on $\mathcal{E}_t$, and at most $d - m - |S^* \setminus \hat{S}_t^{\mathcal{L}}|$ suboptimal arms could have higher UCB index (each requiring its confidence interval to inflate above $\Delta_{\min}$), the hedge set contains at least one optimal arm when:

$$h_t \ge |\{i \notin S^* \cup \hat{S}_t^{\mathcal{L}} : \hat{\mu}_{i,t} + \beta_{i,t} \ge \mu_{j^*}\}| + 1.$$

The number of such "over-estimated" suboptimal arms is at most $d \cdot \exp(-N_{i,t}\Delta_{\min}^2/2)$ in expectation, which is $o(1)$ after sufficient samples. Thus $h_t \ge 1$ suffices for large t, and $h_t = \lceil h_{\max}(1 - \tau_t) \rceil \ge 1$ whenever $\tau_t < 1$, which holds whenever $\varepsilon > 0$ triggers any disagreement in consistency or validation. □

B Proof of Theorem 3 (Lower Bound)

Proof. **Construction.** Let $d = 2m$ and consider two instances:

- P_1: Arms $\{1, \ldots, m\}$ have mean $1/2 + \Delta$; arms $\{m+1, \ldots, 2m\}$ have mean $1/2$. The CLO returns $\{1, \ldots, m\}$ w.p. $1 - \varepsilon$ and $\{m+1, \ldots, 2m\}$ w.p. ε.
- P_2: Arms $\{m+1, \ldots, 2m\}$ have mean $1/2 + \Delta$; arms $\{1, \ldots, m\}$ have mean $1/2$. The CLO behaves identically: returns $\{1, \ldots, m\}$ w.p. $1 - \varepsilon$ and $\{m+1, \ldots, 2m\}$ w.p. ε.

On P_1, the CLO is ε-corrupted (correct w.p. $1 - \varepsilon$). On P_2, the CLO is $(1 - \varepsilon)$-corrupted (correct w.p. ε). But the CLO's *behavior* is identical on both instances.

Indistinguishability. Any algorithm that plays $\{1, \ldots, m\}$ at round t collects reward information only about those arms. To distinguish P_1 from P_2, the algorithm must play arms from $\{m+1, \ldots, 2m\}$ sufficiently often. The CLO suggests $\{m+1, \ldots, 2m\}$ only with probability ε per round, so on average the algorithm receives $O(\varepsilon T)$ rounds of information about the second group when following the oracle.

By a change-of-measure argument (Lemma 15.1 of Lattimore and Szepesvári [11]), distinguishing P_1 from P_2 using N samples from each contested arm requires $N = \Omega(1/\Delta^2)$. With m contested arms each sampled $O(\varepsilon T/m)$ times (when following the oracle), the algorithm can distinguish only after $T = \Omega(m/(\varepsilon\Delta^2))$ rounds.

Regret accounting. On P_2, any algorithm that predominantly plays $\{1, \ldots, m\}$ (as the oracle suggests) incurs per-round regret $m\Delta$. Until the algorithm has enough evidence to override the oracle, it incurs regret $\Omega(m\Delta \cdot \min(T, m/(\varepsilon\Delta^2)))$. Setting $\Delta = \Delta_{\min}$:

$$\mathbb{E}[R_T] \ge \Omega\left(\min\left(m\Delta_{\min}T,\ \frac{m^2}{\varepsilon\Delta_{\min}}\right)\right) = \Omega\left(\frac{\varepsilon m T}{\Delta_{\min}} \cdot \frac{m\Delta_{\min}^2}{\varepsilon}\right).$$

Optimizing over Δ, the per-round cost of not knowing ε is $\Omega(\varepsilon m/\Delta_{\min})$, giving total regret

$\Omega(\varepsilon m \log T/\Delta_{\min})$. □

C ELLM Failure Under Consistent Corruption

We construct a counter-example showing that ELLM-style exploration bonuses fail under consistently wrong LLM guidance, while LLM-CUCB-AT's posterior validation provides protection.

Setup. $d = 20$, $m = 5$, arms $\{1, \ldots, 5\}$ are optimal (mean 0.8), arms $\{6, \ldots, 10\}$ are "plausible but suboptimal" (mean 0.6), remaining arms have mean 0.3. The LLM deterministically suggests $\{6, \ldots, 10\}$ at every round ($\kappa = 1$, $\varepsilon = 1$).

ELLM behavior. ELLM uses the LLM suggestion to define exploration bonuses on the suggested arms. Since the LLM consistently suggests $\{6, \ldots, 10\}$, ELLM repeatedly boosts these arms' exploration scores, pulling exploration *toward* the suboptimal set. Empirically, ELLM incurs $\Omega(T)$ regret because the exploration bonus reinforces the LLM's mistake.

LLM-CUCB-AT behavior. After $O(1/\Delta_{\text{set}}^2) = O(1/(0.2)^2) = O(25)$ rounds, the posterior validation score ρ_t drops below 1 (since $\sum_{i=6}^{10} \hat{\mu}_i \approx 3.0 < \sum_{i=1}^{5} \hat{\mu}_i \approx 4.0$). The composite trust $\tau_t < 1$ triggers a hedge, and the UCB-ranked hedge set includes arms from $\{1, \ldots, 5\}$. Within $O(d \log d)$ additional rounds, the algorithm converges to playing $\{1, \ldots, 5\}$.

D Warm-Start Thompson Sampling: Worst-Case Analysis

Proposition 2. *There exists a problem instance where Warm-Start CTS with adversarial LLM priors incurs $\Omega(\sqrt{T/m})$ additional regret compared to standard CTS, whereas LLM-CUCB-AT detects the adversarial prior in $O(1/\Delta_{\min}^2)$ rounds.*

Proof sketch. Consider the instance from Appendix C. If the LLM assigns score 10 to arms $\{6, \ldots, 10\}$ (mapping to Beta$(10, 0)$, i.e., near-certain belief that these are optimal) and score 0 to arms $\{1, \ldots, 5\}$, Warm-Start CTS requires $\Omega(10/\Delta^2)$ samples of each arm to overcome the adversarial prior, during which it incurs $O(m\Delta)$ regret per round. The posterior update speed depends on the prior strength: stronger priors require more evidence to overcome. In contrast, LLM-CUCB-AT's trust score does not "believe" the prior—it verifies it against UCB estimates, which concentrate in $O(\log(t)/\Delta^2)$ rounds regardless of any prior. □

E Extension to ϕ-Mixing Re-Queries

When re-query outputs are not independent but instead form a ϕ-mixing sequence with mixing coefficient ϕ, the consistency estimator's concentration degrades.

Proposition 3. *Under ϕ-mixing with coefficient $\phi < 1$, the consistency score κ_t satisfies:*

$$\mathbb{P}(|\kappa_t - \kappa^*| > \eta) \leq 2 \exp\left(-\frac{K\eta^2}{2(1 + 2\phi/(1-\phi))}\right),$$

where $\kappa^* = \mathbb{E}[\kappa_t]$. For $K = 3$ and $\phi = 0.3$, the effective sample size is $K_{\text{eff}} = K/(1+2\phi/(1-\phi)) \approx 1.6$.

This degradation is absorbed by the posterior validation component of the composite trust, which does not depend on re-query independence.

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